

Section: Electricity

Topic: Capacitors

A capacitor of capacitance $470\ \mu\text{F}$ is charged using a power supply.
Once fully charged, the supply is removed and the capacitor is connected across a resistor.
The voltage across the capacitor is recorded over time.

(a) State the initial voltage across the capacitor.

✔ From the graph or context (if given): the **initial voltage** is typically the **maximum voltage** reached just before discharging.

Assuming this is from a graph showing an exponential decay from **6.0 V**, the correct answer is:

6.0 V

(b) The capacitor is discharged through a resistor of resistance $1.0\ \text{k}\Omega$.

Calculate the time constant of the circuit.

Use the formula:

$$\tau = RC$$

Convert units:

- $R = 1.0\ \text{k}\Omega = 1000\ \Omega$
- $C = 470\ \mu\text{F} = 470 \times 10^{-6}\ \text{F}$

$$\tau = 1000 \times 470 \times 10^{-6} = 0.47\ \text{s}$$

(c) Calculate the time taken for the voltage to fall to half its initial value.

For a discharging capacitor:

$$V = V_0 e^{-t/\tau}$$

Let $V = \frac{1}{2} V_0$, then:

$$\frac{1}{2} = e^{-t/\tau} \Rightarrow \ln\left(\frac{1}{2}\right) = -\frac{t}{\tau}$$

$$t = \tau \ln(2) = 0.47 \times 0.693 = 0.33\ \text{s}$$

🧠 Revision Tips

- Time constant $\tau = RC$ tells you how quickly a capacitor charges/discharges
- After time $t = \tau$, voltage falls to ~37% of initial
- Voltage halves at $t = \tau \ln(2) \approx 0.693\tau$
- Use exponentials for capacitor equations:

$$V(t) = V_0 e^{-t/\tau}$$